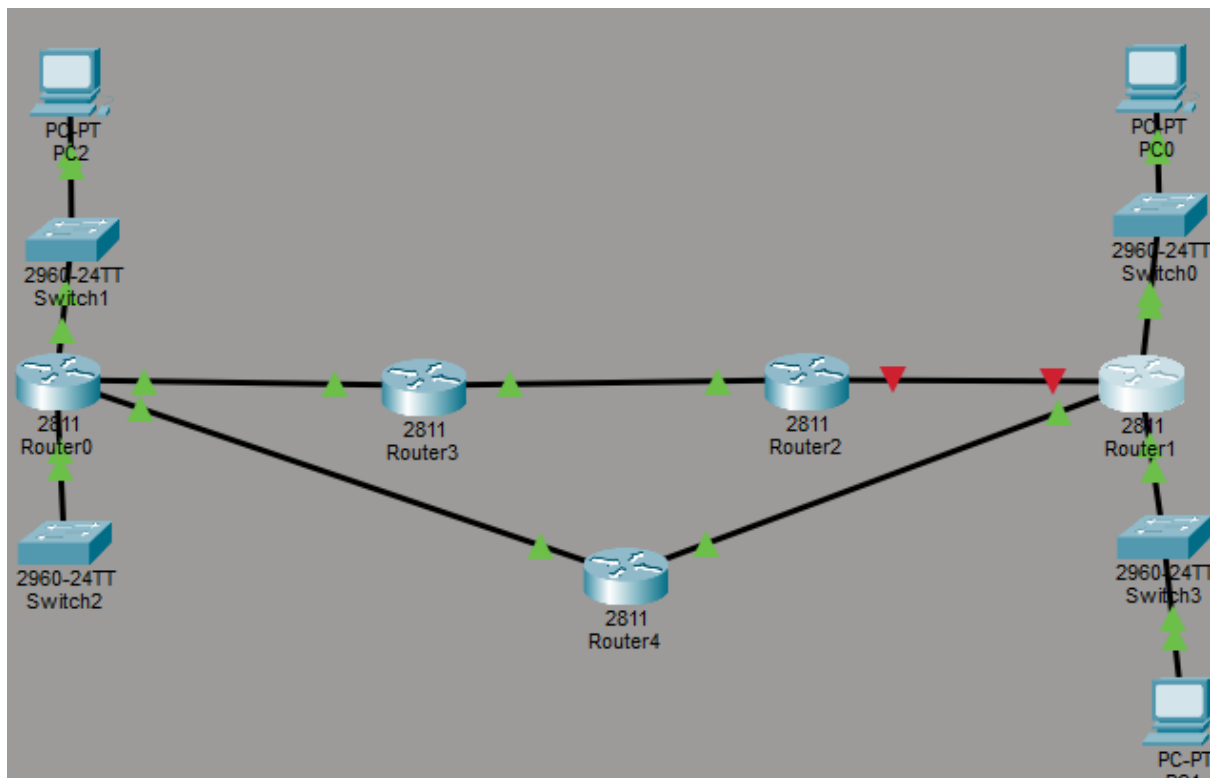


Week 8: screenshots and reflection questions



1.) Configuring RIPv1 on every router:

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 10.0.0.0
Router(config-router)#no auto-summary
Router(config-router)#end
```

2.) Observing RIPv1 updates:

```
Router#debug ip rip
RIP protocol debugging is on
Router#
```

3.) Upgrading to RIPv2

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#end
Router#
```

4.) Observing RIPv2 updates:

```

Router#debug ip rip
RIP protocol debugging is on
Router#RIP: sending v2 update to 224.0.0.9 via FastEthernet0/0 (10.0.0.1)
RIP: build update entries
    10.0.1.0/24 via 0.0.0.0, metric 1, tag 0
    10.0.2.0/24 via 0.0.0.0, metric 1, tag 0
    10.0.3.0/24 via 0.0.0.0, metric 1, tag 0
RIP: sending v2 update to 224.0.0.9 via FastEthernet0/1 (10.0.1.1)
RIP: build update entries
    10.0.0.0/24 via 0.0.0.0, metric 1, tag 0
    10.0.2.0/24 via 0.0.0.0, metric 1, tag 0
    10.0.3.0/24 via 0.0.0.0, metric 1, tag 0
RIP: sending v2 update to 224.0.0.9 via FastEthernet1/0 (10.0.2.1)
RIP: build update entries
    10.0.0.0/24 via 0.0.0.0, metric 1, tag 0
    10.0.1.0/24 via 0.0.0.0, metric 1, tag 0
    10.0.3.0/24 via 0.0.0.0, metric 1, tag 0
RIP: sending v2 update to 224.0.0.9 via FastEthernet1/1 (10.0.3.1)
RIP: build update entries
    10.0.0.0/24 via 0.0.0.0, metric 1, tag 0
    10.0.1.0/24 via 0.0.0.0, metric 1, tag 0
    10.0.2.0/24 via 0.0.0.0, metric 1, tag 0

```

5.) Comparing router protocols:

```

Router#show ip rip database
10.0.0.0/24    auto-summary
10.0.0.0/24    directly connected, FastEthernet0/0
10.0.1.0/24    auto-summary
10.0.1.0/24    directly connected, FastEthernet0/1
10.0.2.0/24    auto-summary
10.0.2.0/24    directly connected, FastEthernet1/0
10.0.3.0/24    auto-summary
10.0.3.0/24    directly connected, FastEthernet1/1
Router#

```

6.) Configuring OSPF:

```

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG I: Configured from console by console

```

7.) Checking R1 routing table:

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       10.0.0.0/24 is directly connected, FastEthernet0/0
L       10.0.0.1/32 is directly connected, FastEthernet0/0
C       10.0.1.0/24 is directly connected, FastEthernet0/1
L       10.0.1.1/32 is directly connected, FastEthernet0/1
C       10.0.2.0/24 is directly connected, FastEthernet1/0
L       10.0.2.1/32 is directly connected, FastEthernet1/0
C       10.0.3.0/24 is directly connected, FastEthernet1/1
L       10.0.3.1/32 is directly connected, FastEthernet1/1

```

8.) Disabling fastethernet 0/0:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet 0/0
Router(config-if)#shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down

Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

9.) OSPF database on R1:

```
Router#show ip ospf database
      OSPF Router with ID (10.0.3.1) (Process ID 1)

      Router Link States (Area 0)

Link ID      ADV Router   Age         Seq#          Checksum Link count
10.0.3.1     10.0.3.1     184         0x80000005   0x00e40a 3
```

Reflection questions:

- 1.) RIPv1 uses broadcast traffic, for RIPv2 it uses multicast traffic.
- 2.) Because RIP has learned that there are two possible paths to the 10.1.1.0/24
- 3.) RIP routes are no longer present on R1 routing table, because OSPF has a lower administrative distance than RIP. The reason why there is only one route to 10.1.1.0 is because routing table selects the preferred route.
- 4.) What changed was outgoing interface, OSPF cost, route metric and route availability
- 5.) RIP database contains routing info learned from neighbouring routers and OSPF database contains Link-State Advertisement describing the network topology, routers and links. The difference exist because RIP is a distance-vector that learns routes from neighbours and OSPF is a link-state protocol that builds a topology database and calculates shortest paths.